

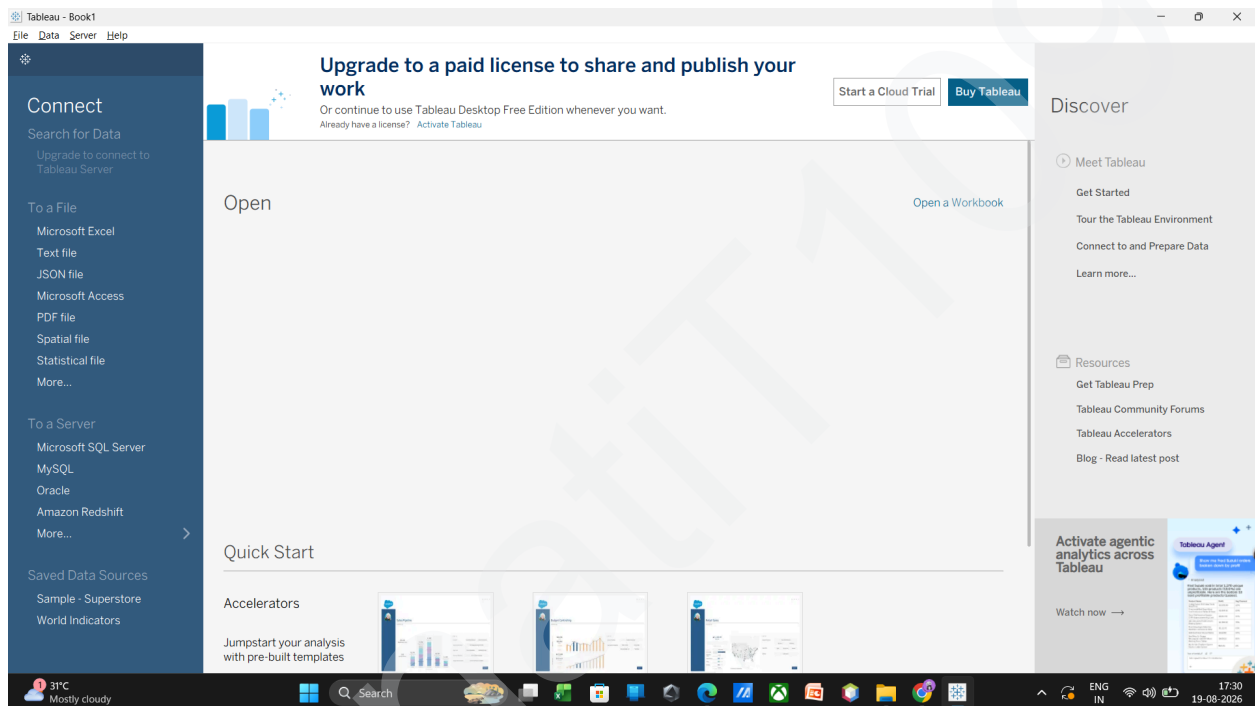
SHETH L.U.J AND SIR M.V COLLEGE

PRACTICAL NO 8

SUBJECT - Data Visualization with Power BI and Tableau

AIM - Performing Advanced Analytics using Trends, Clustering, and Forecasting

- a) Add and interpret trend lines (linear, exponential)
- b) Customize and analyze trend models
- c) Perform clustering on datasets
- d) Analyze distributions using built-in tools
- e) Apply forecasting techniques
- f) Compare different statistical models
- g) Export and interpret statistical results



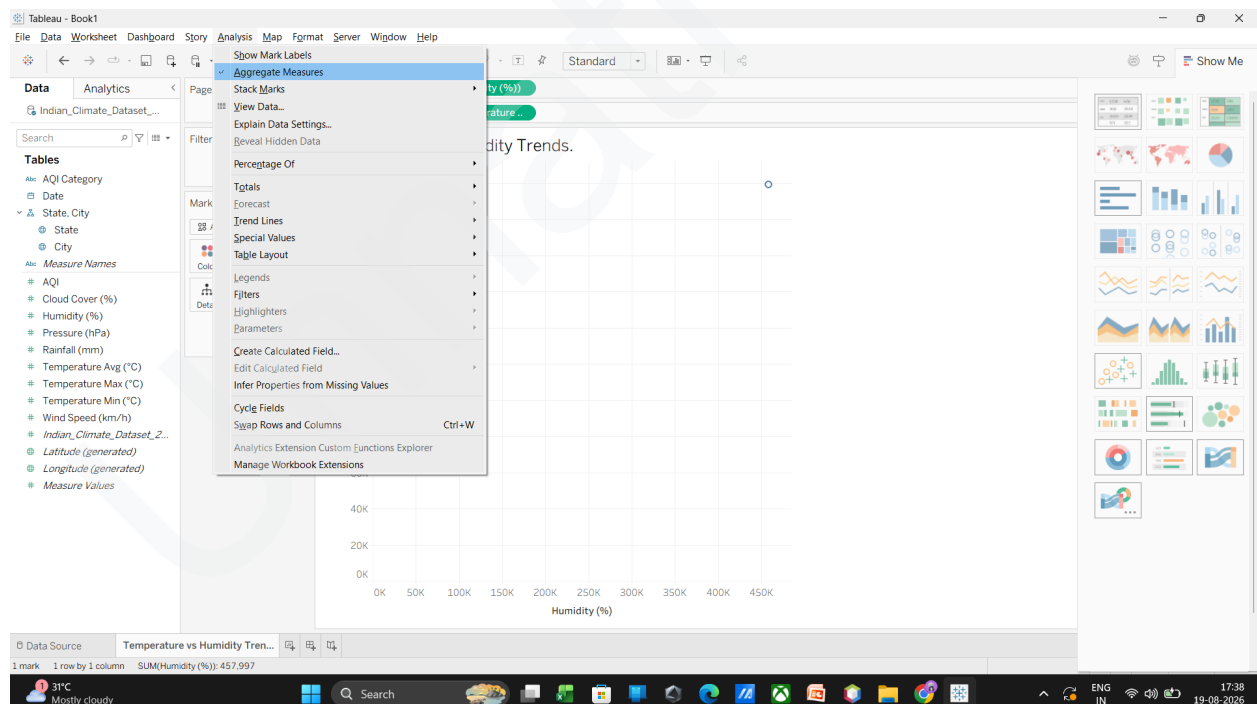
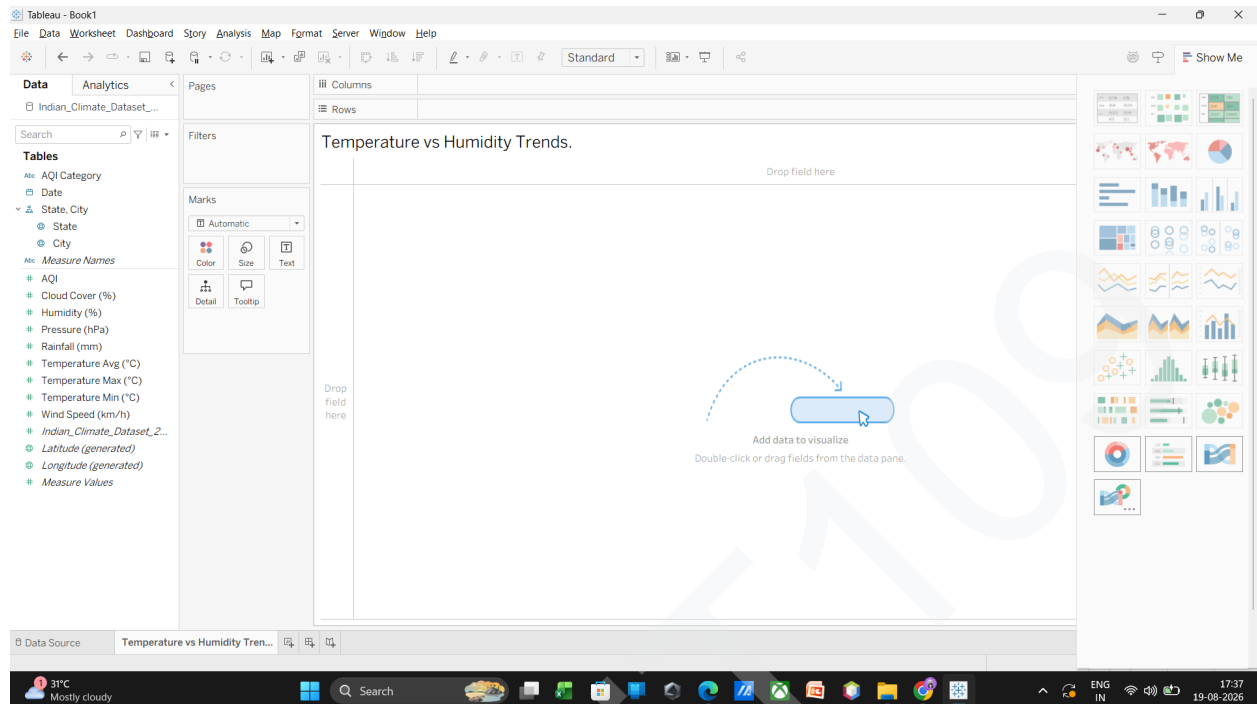
A: Add and Interpret Trend Lines (Linear, Exponential)

1. Connect Dataset & Build a Scatter Plot / Line Chart

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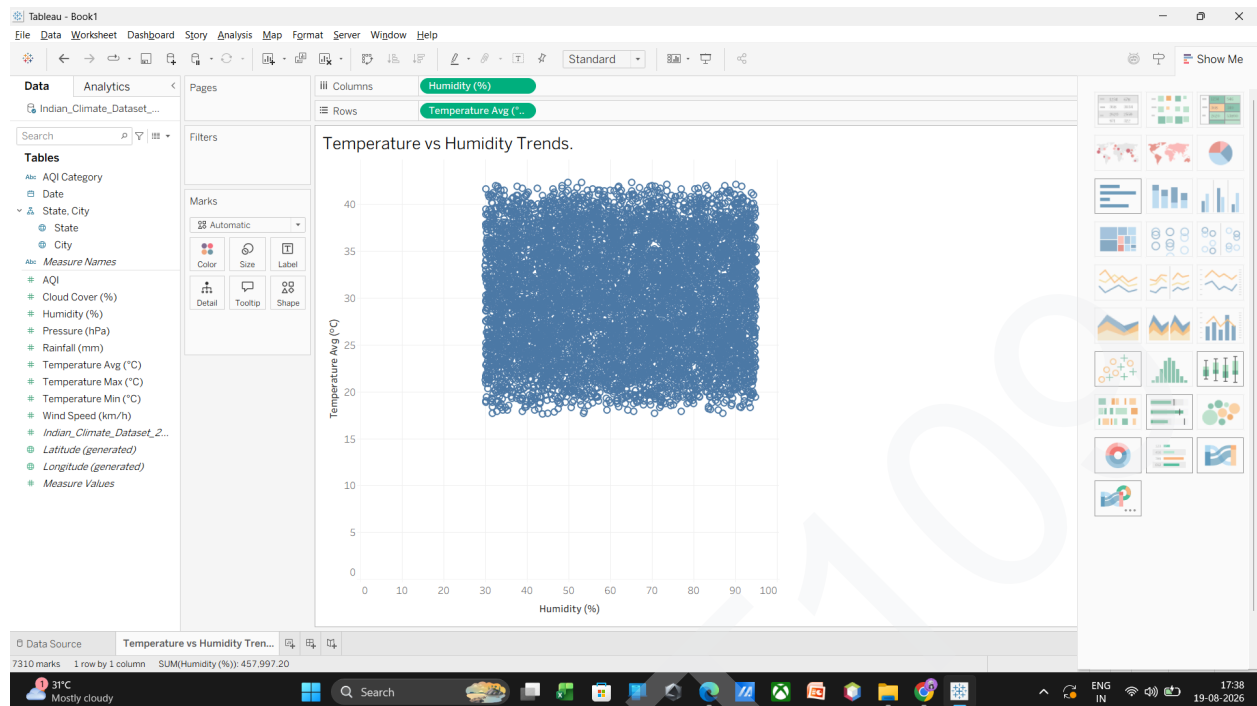
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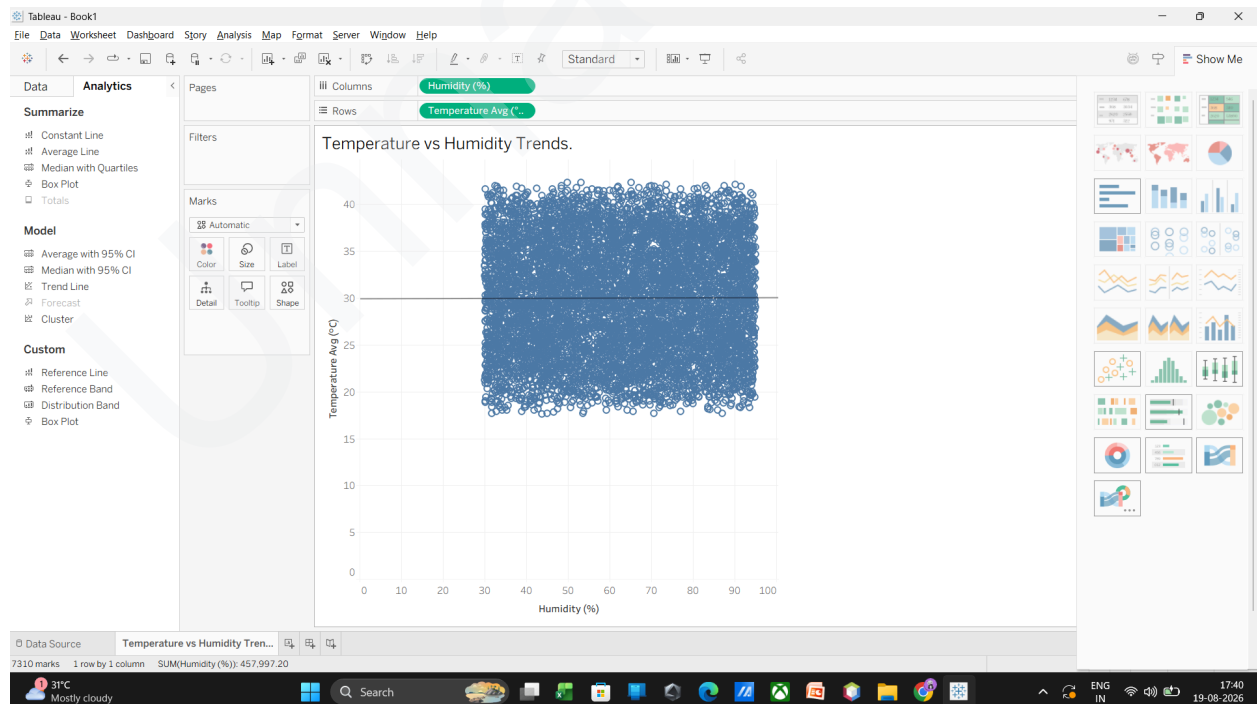


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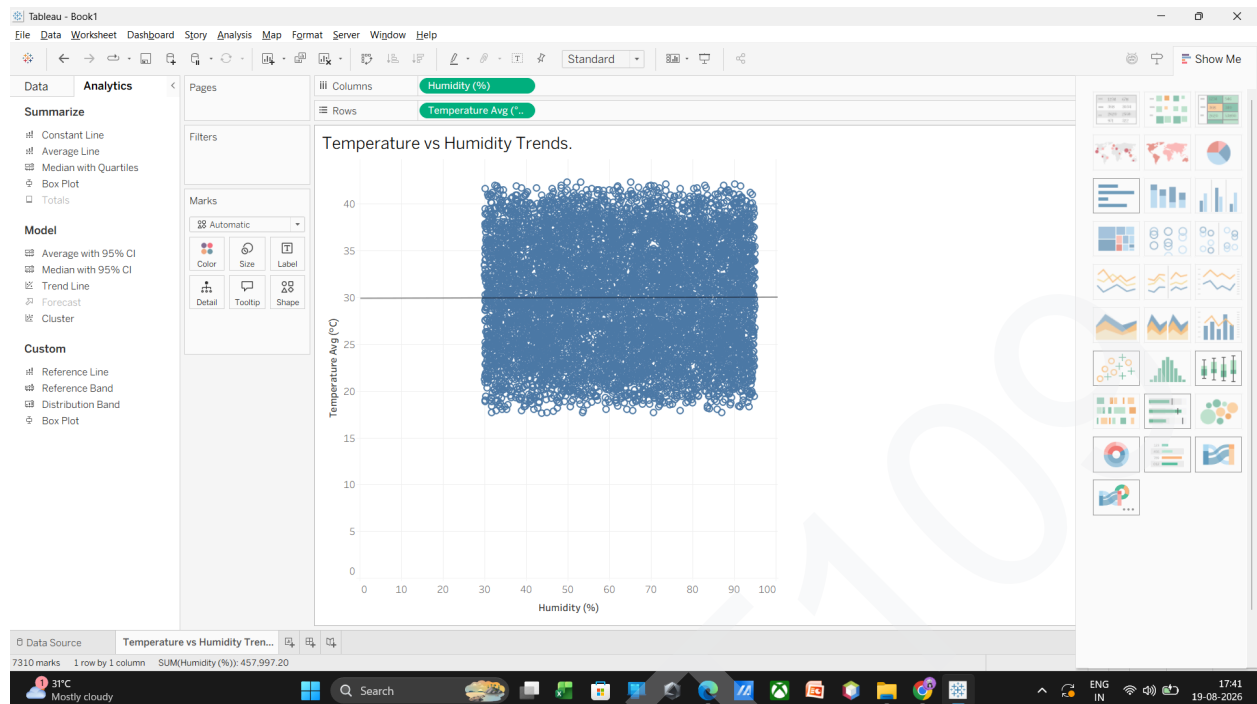


2. Add a Linear Trend Line

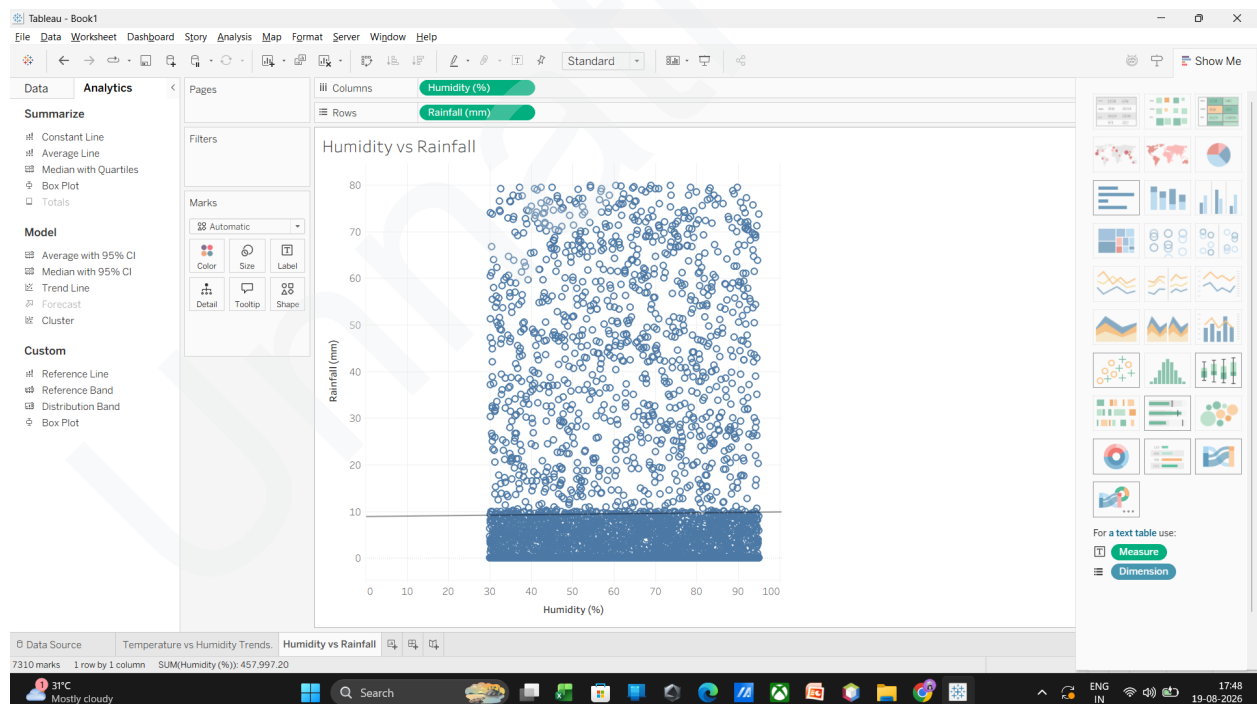


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3. Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)



4. Interpret the Results

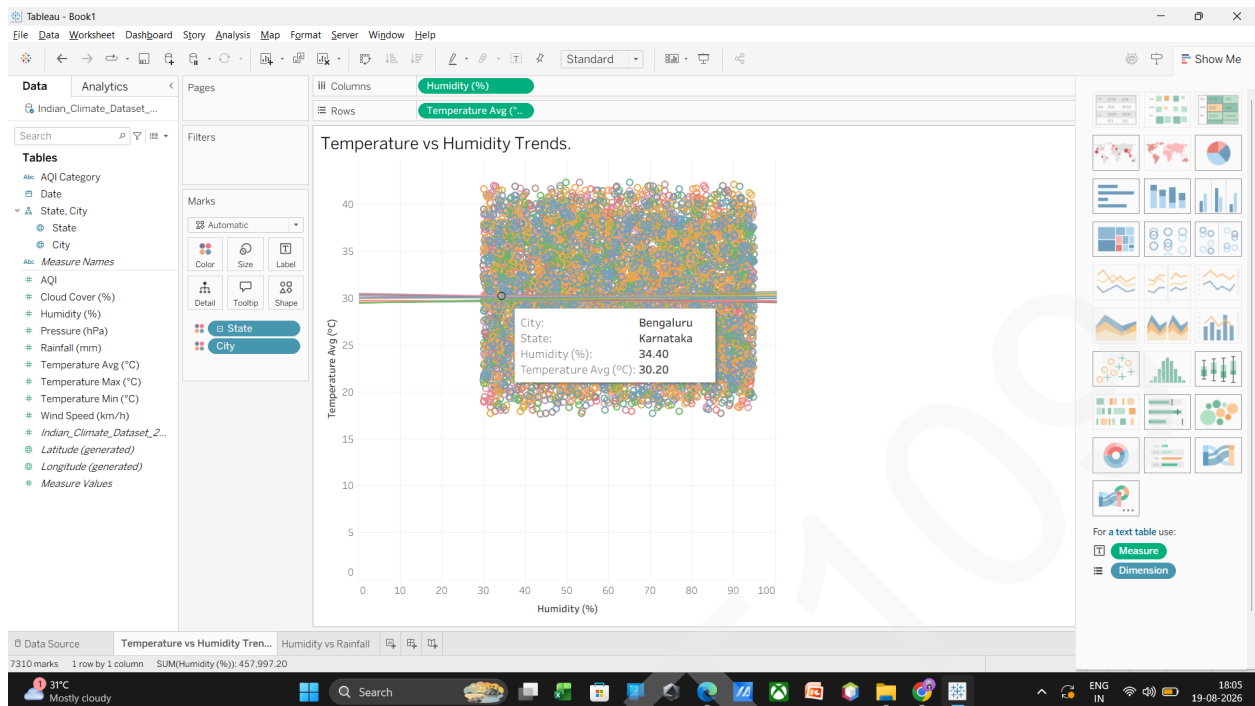
B: Customize and Analyze Trend Models

1. Disaggregate Trend Lines by City

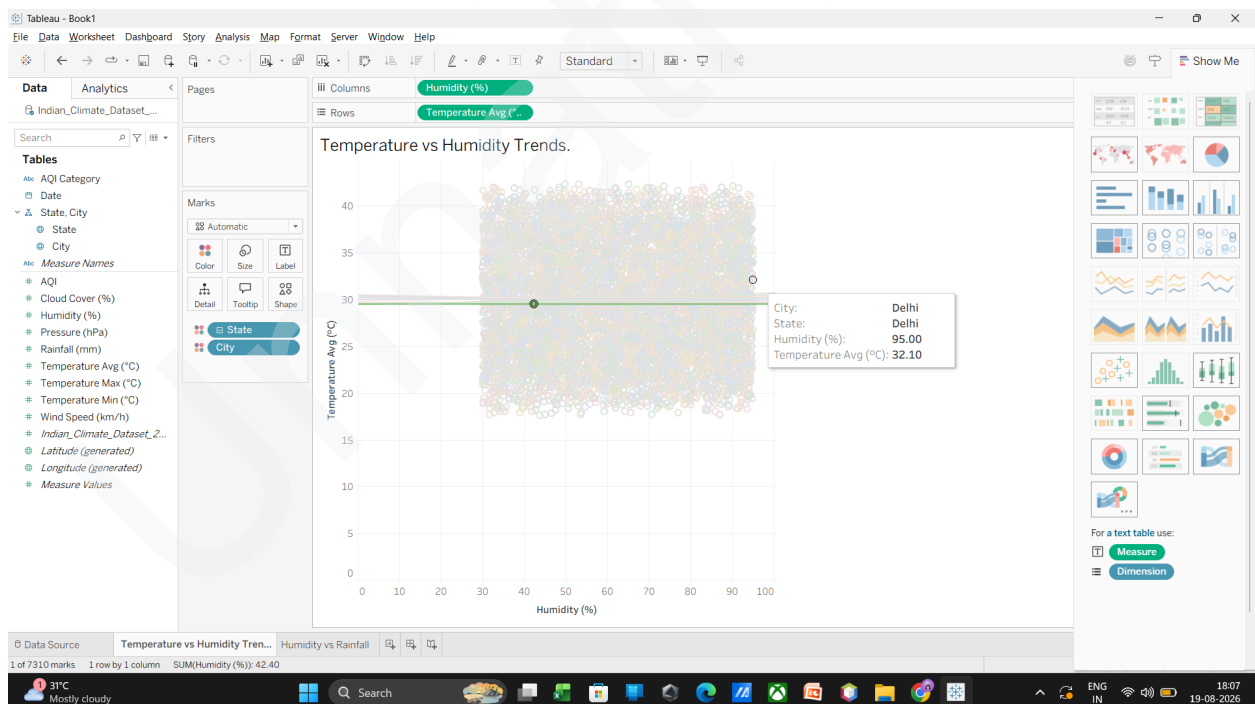
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2 Compare Different Model Types (Polynomial & Logarithmic)



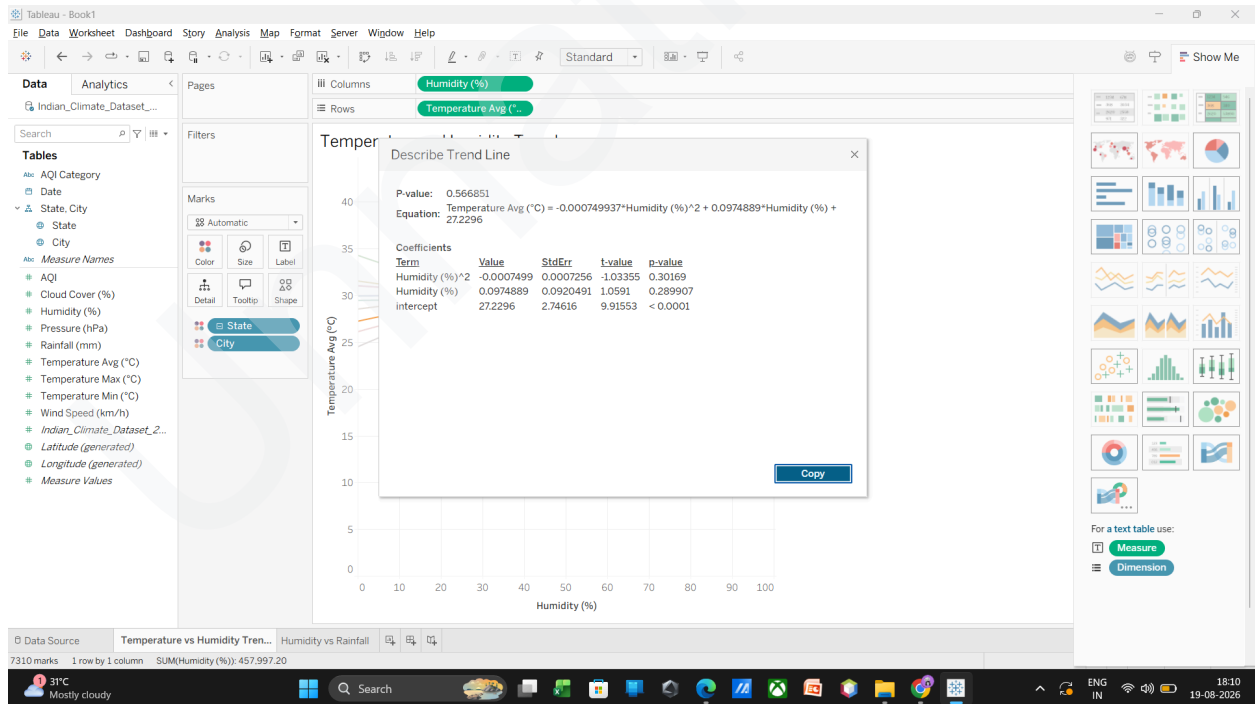
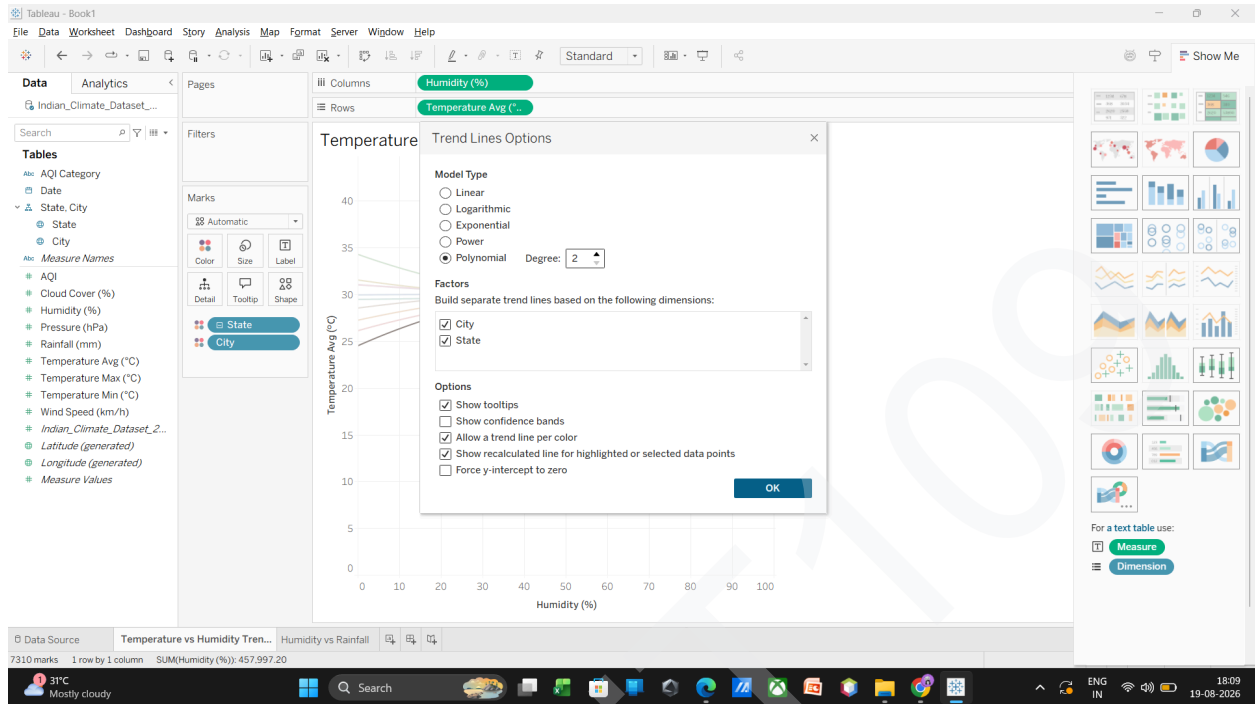
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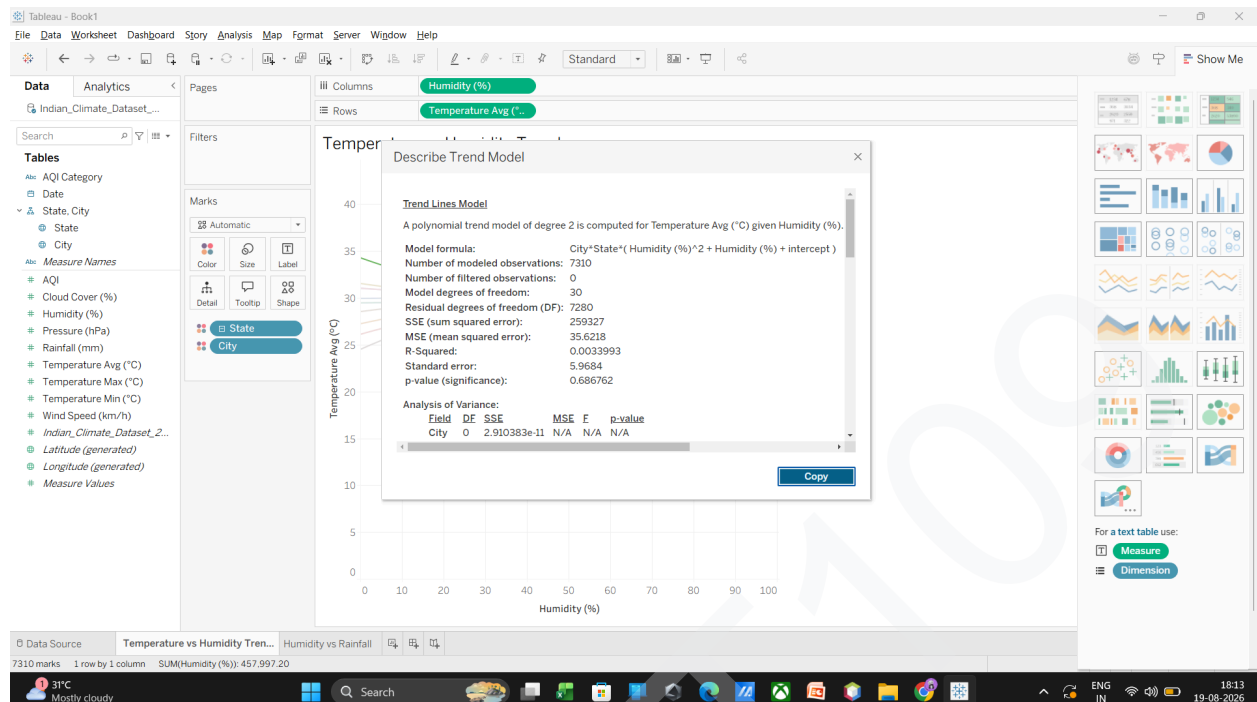
3. Access Full Statistical Summaries (Describe Model)



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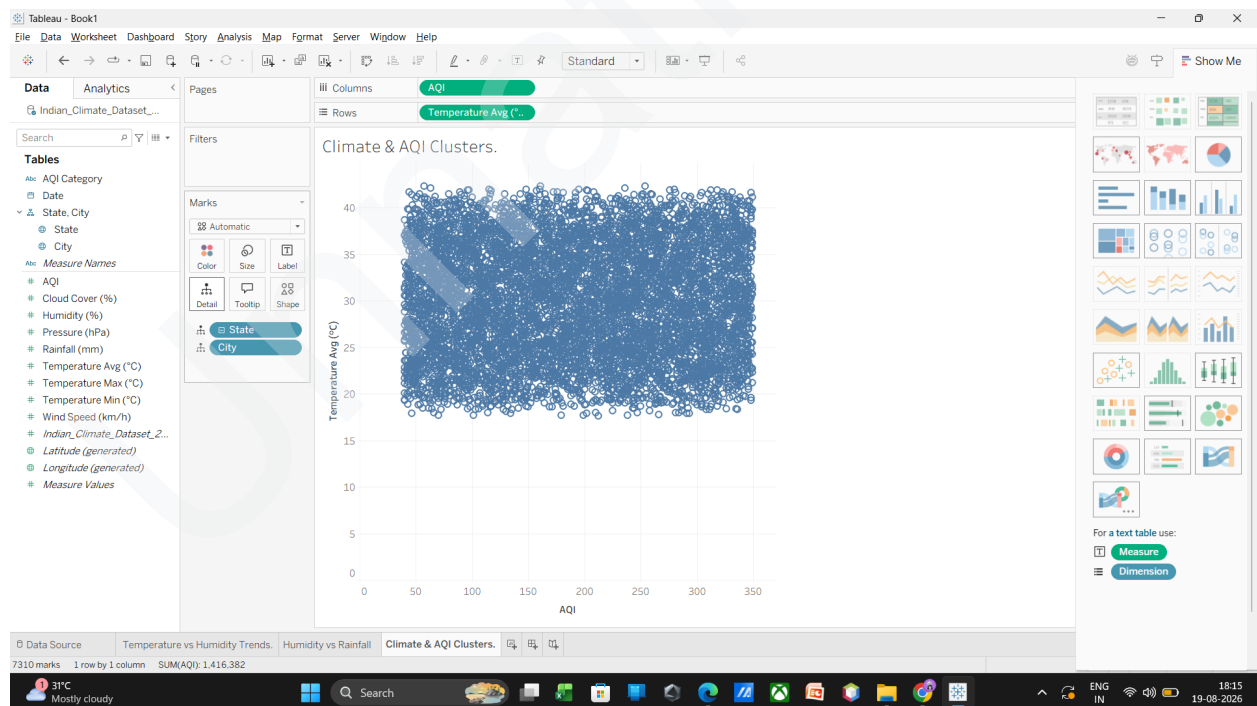
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C: Perform Clustering on Datasets

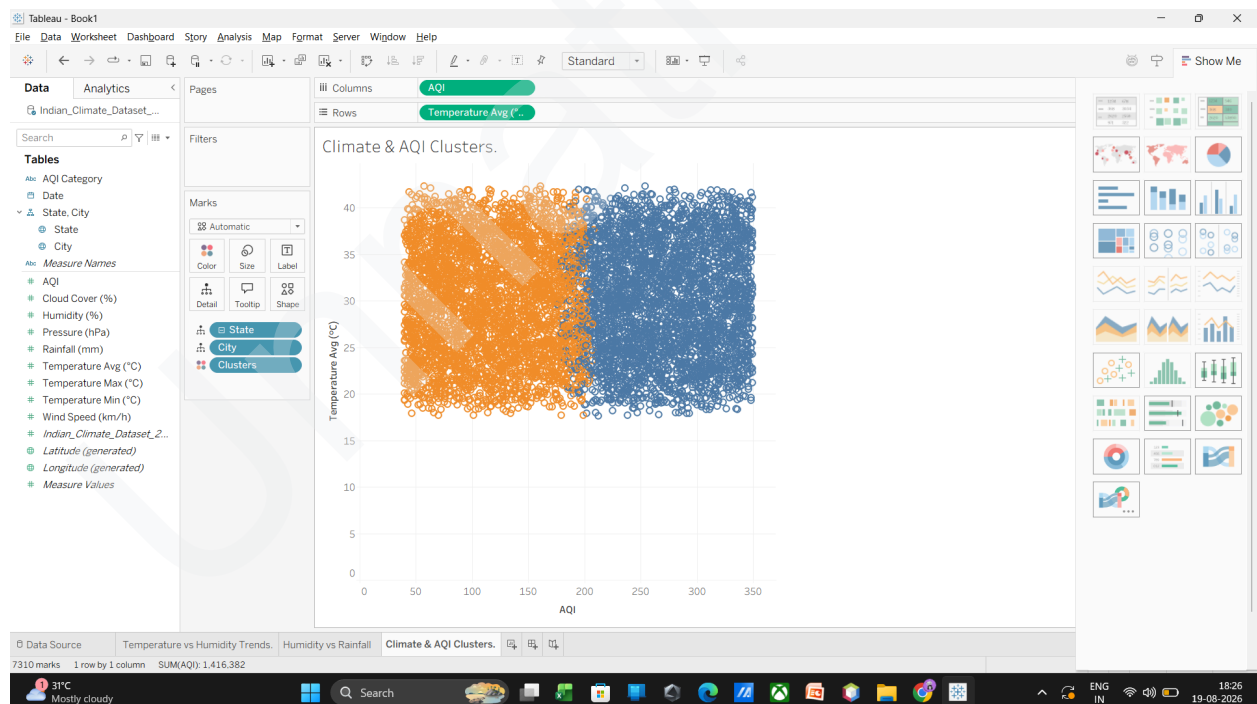
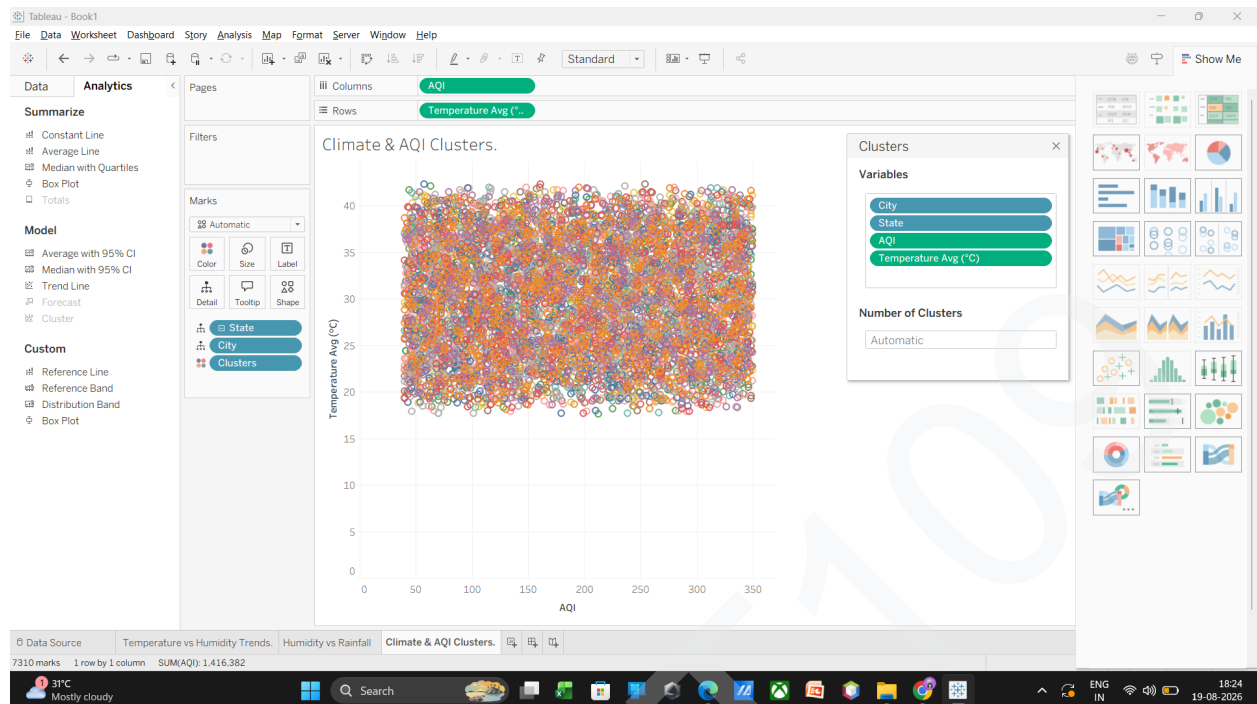
1. Build a Multi-Metric Scatter Plot Matrix



2. Apply Built-in Tableau Clustering

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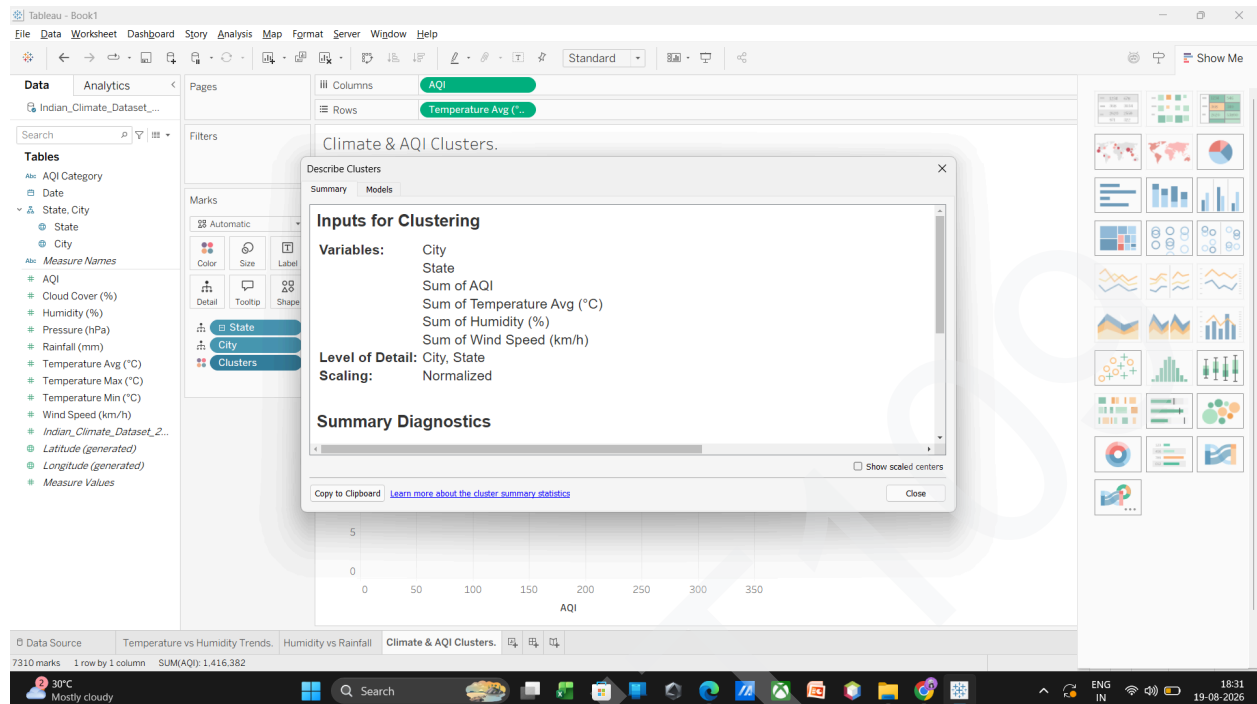
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3. Analyze Cluster Profiles & Save as Dimension

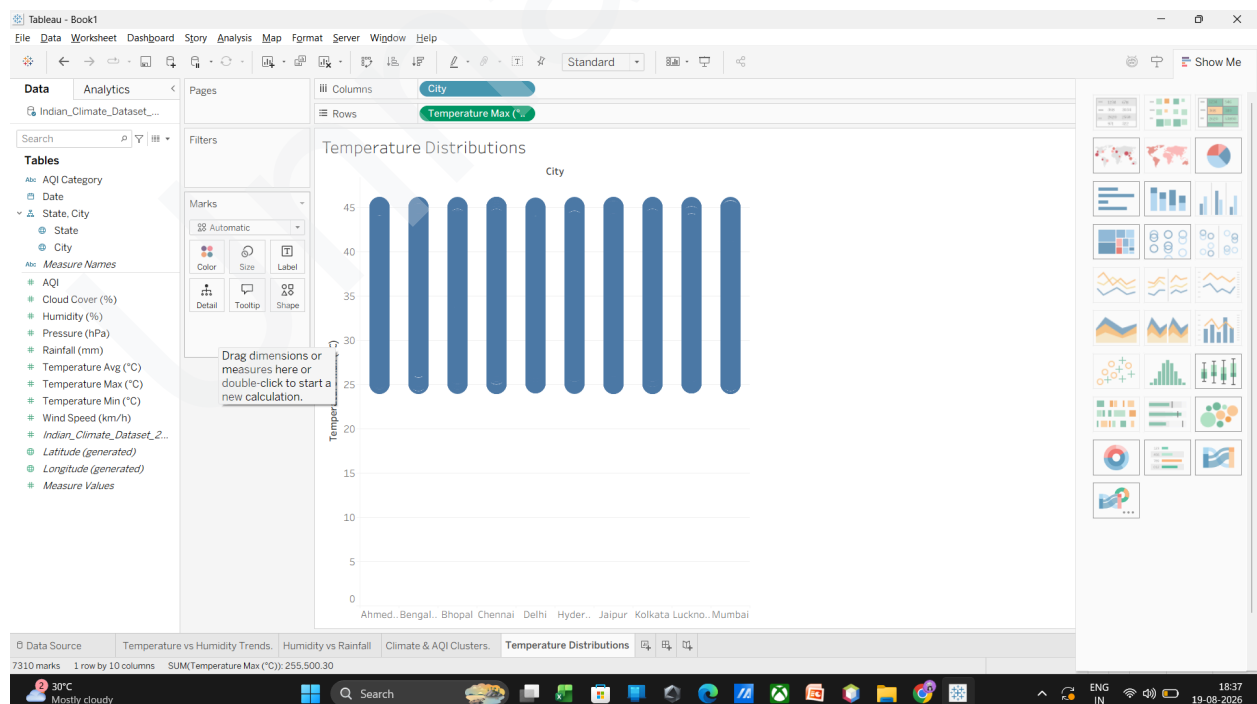
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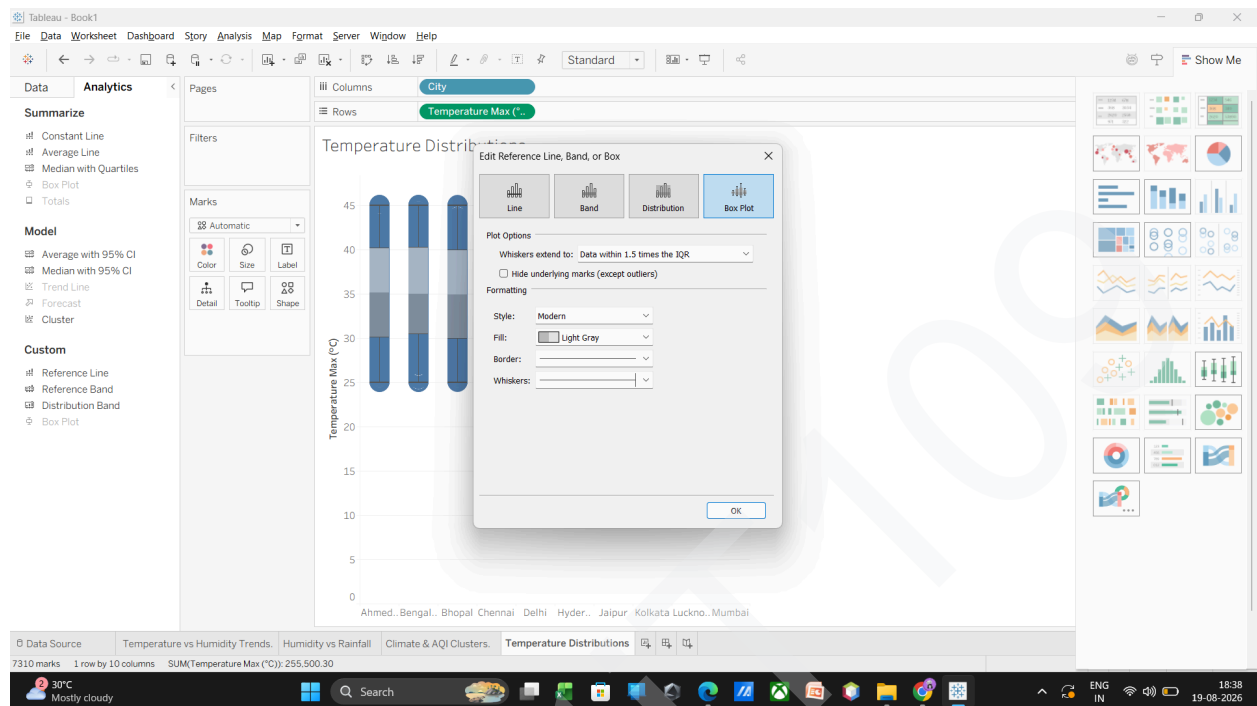
D: Analyze Distributions using Built-in Tools

1. Create a Box Plot for Temperature Distributions by City

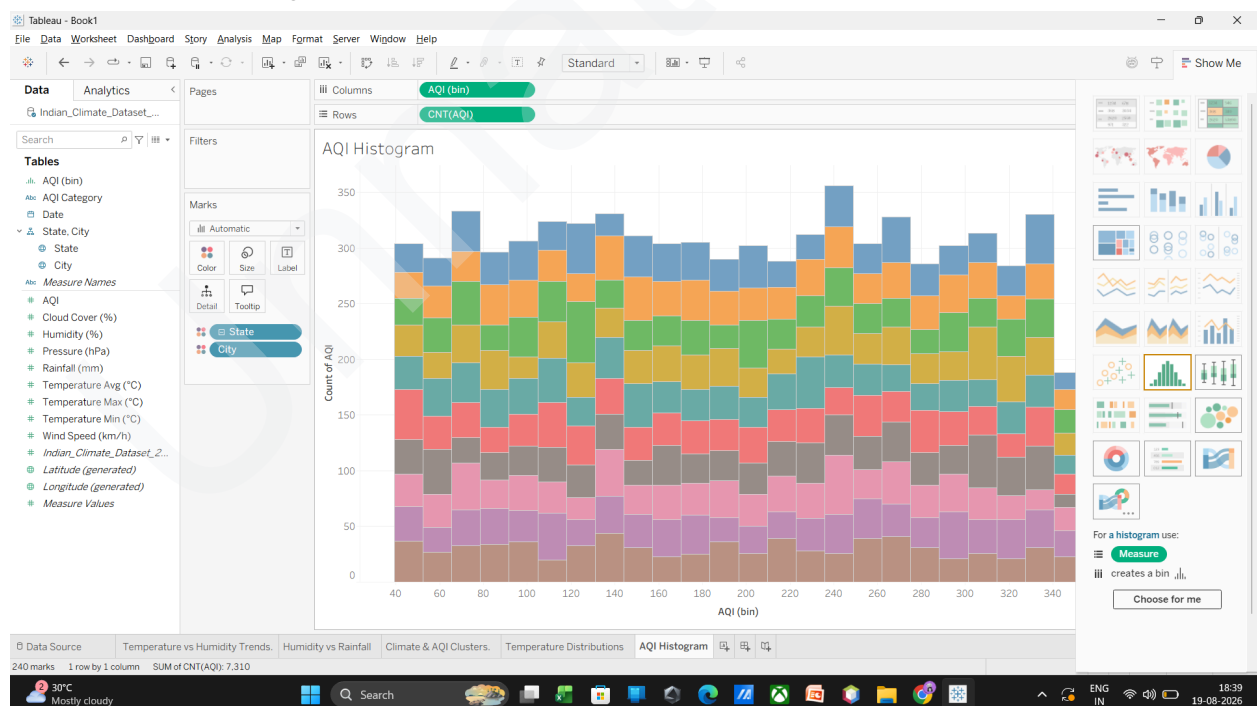


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2. Build an AQI Histogram & Quantile/Percentile View



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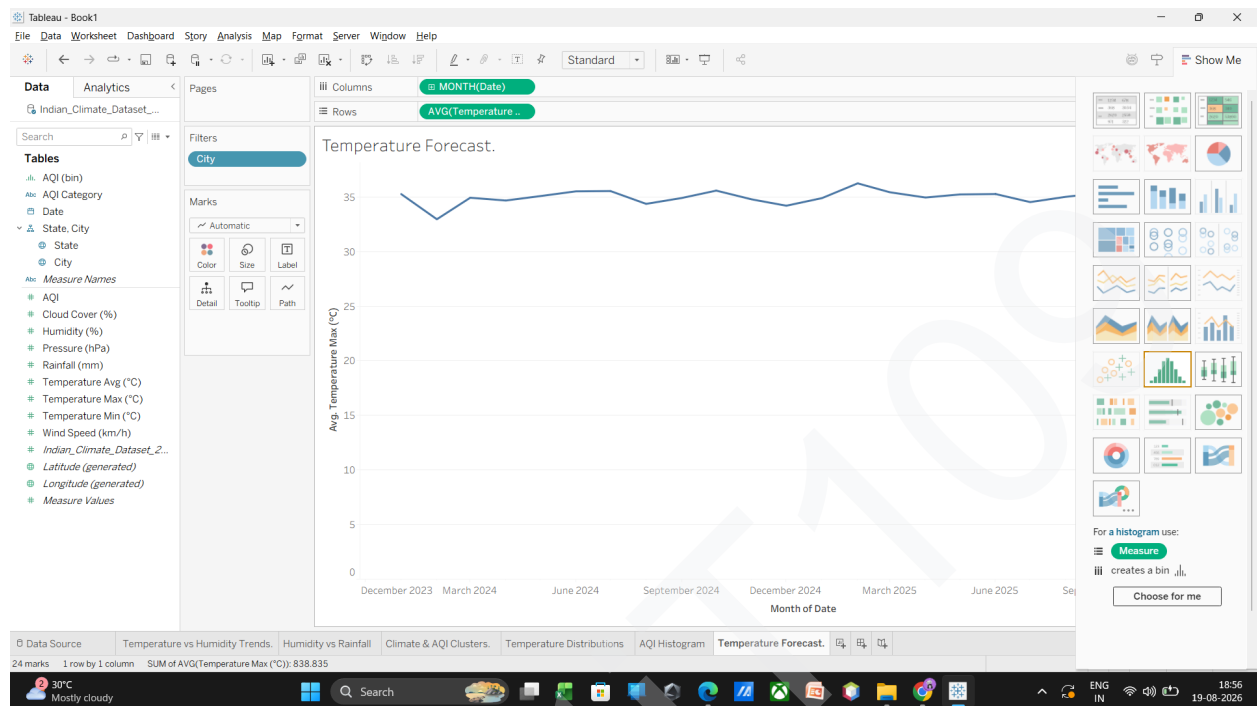
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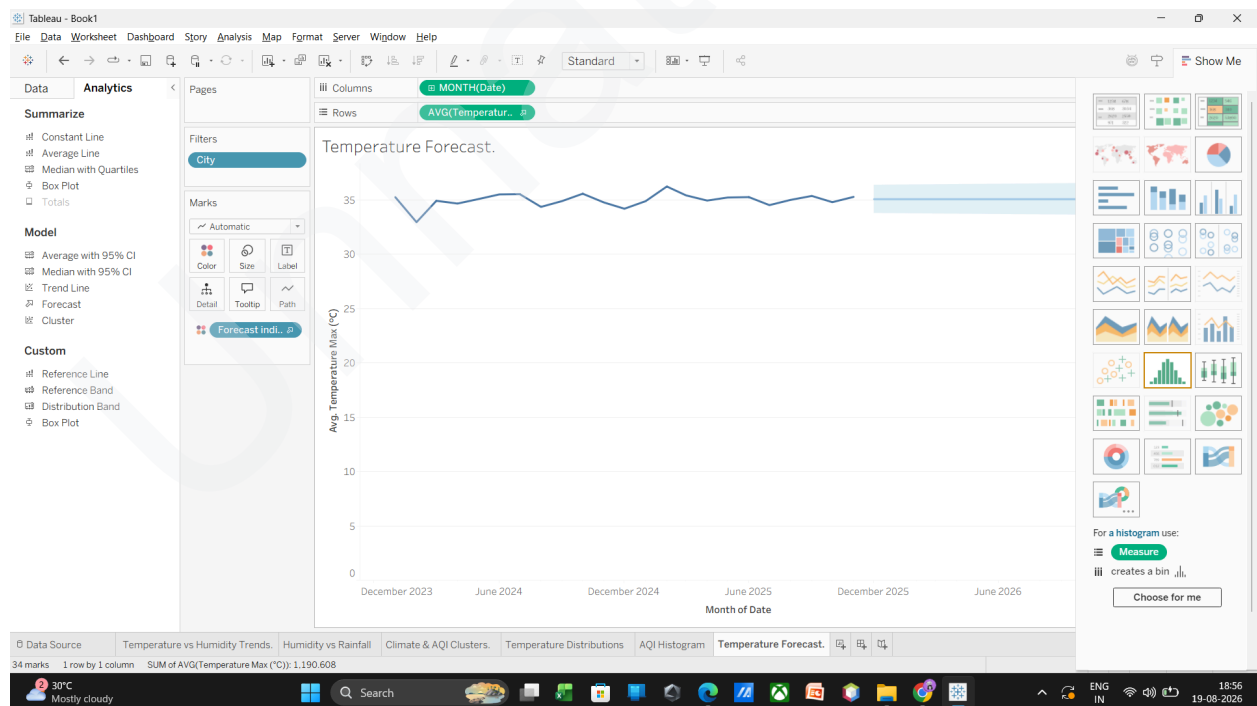
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E: Apply Forecasting Techniques

1. Set Up a Time Series Line Chart



2. Enable Tableau Forecasting

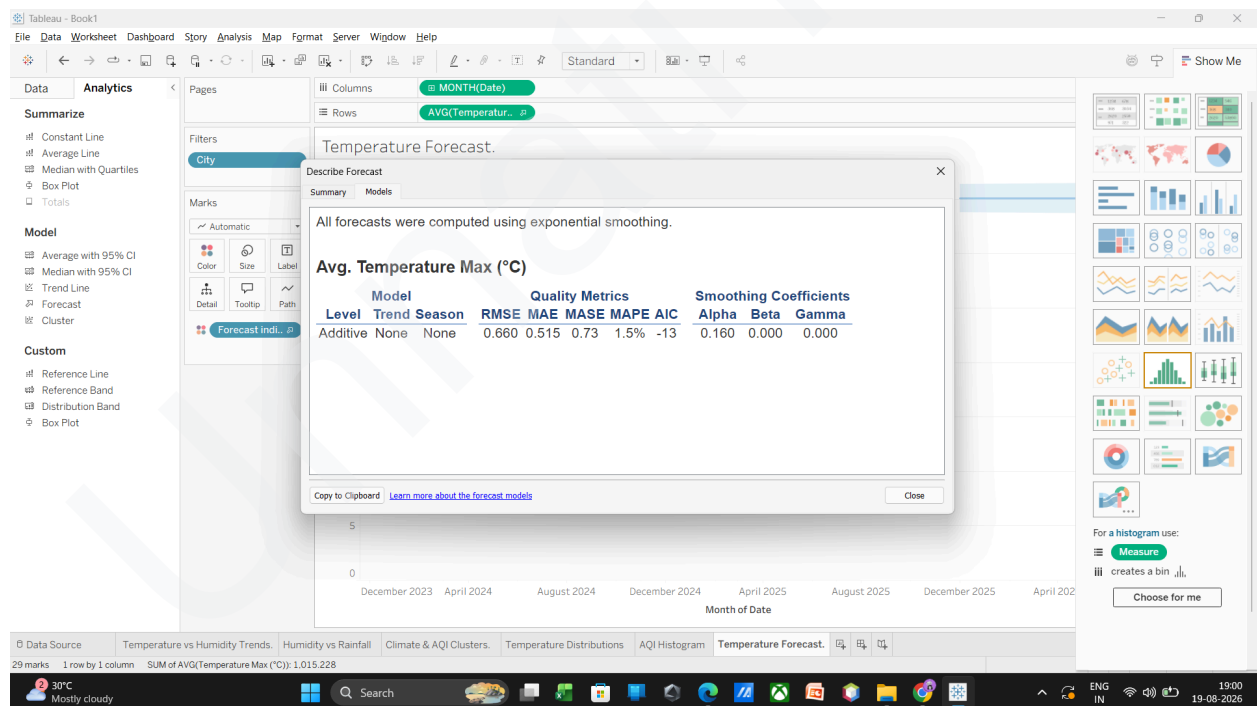
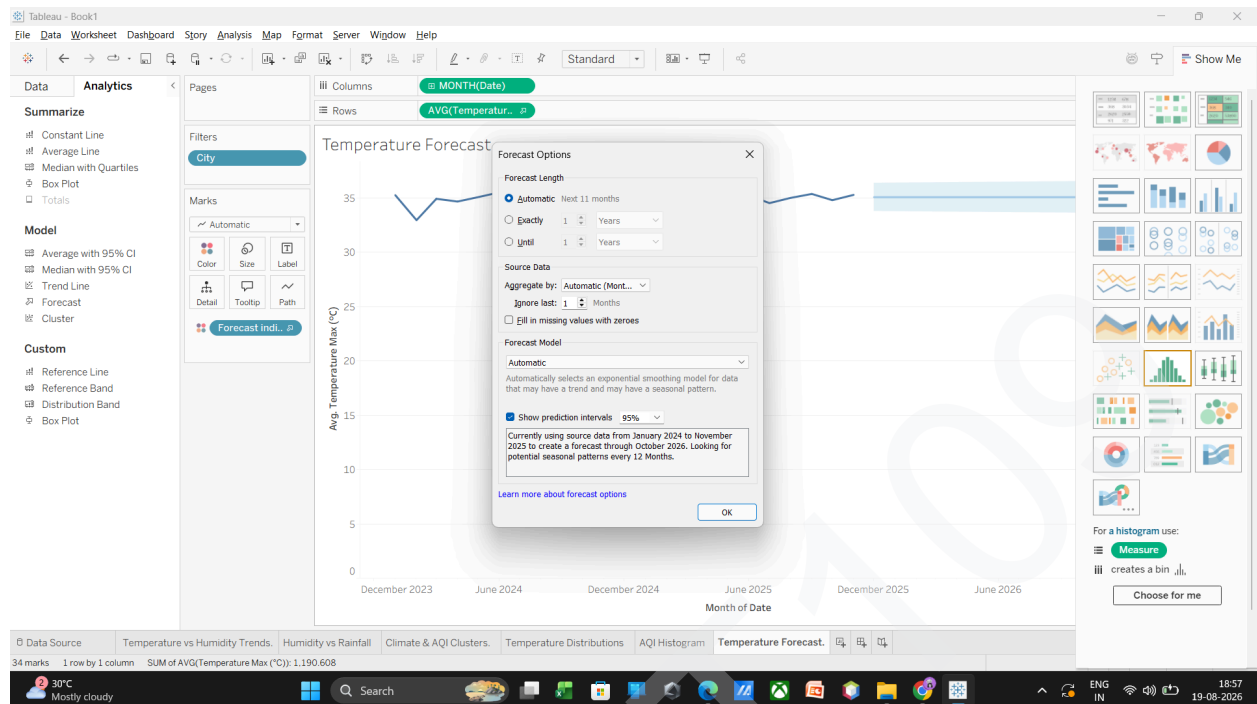


3. Customize & Analyze the Forecast

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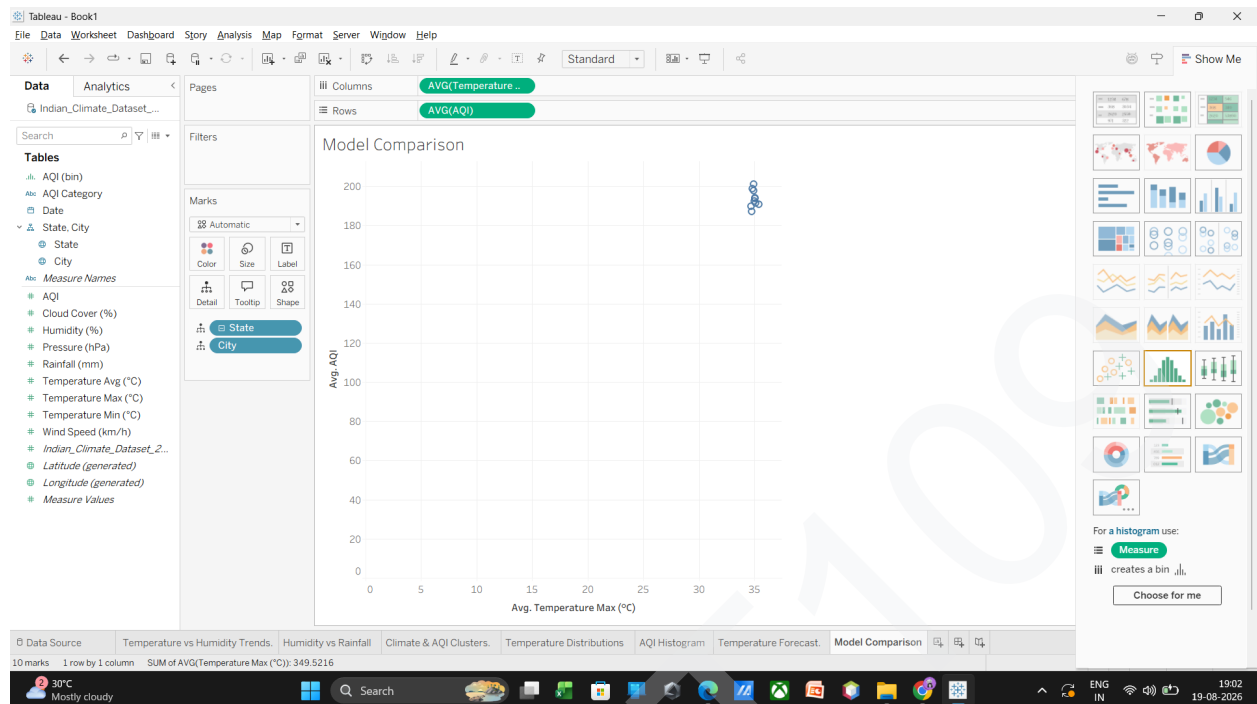


F: Compare different Statistical Models

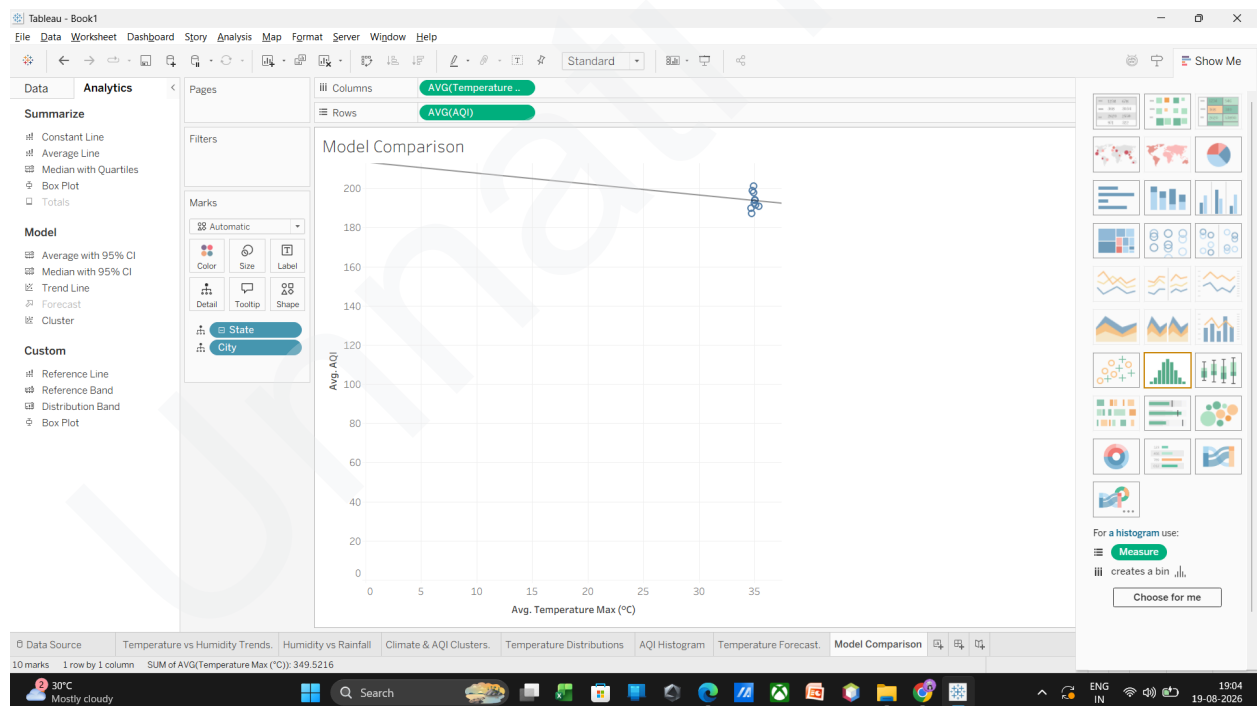
1. Set Up the Comparison View

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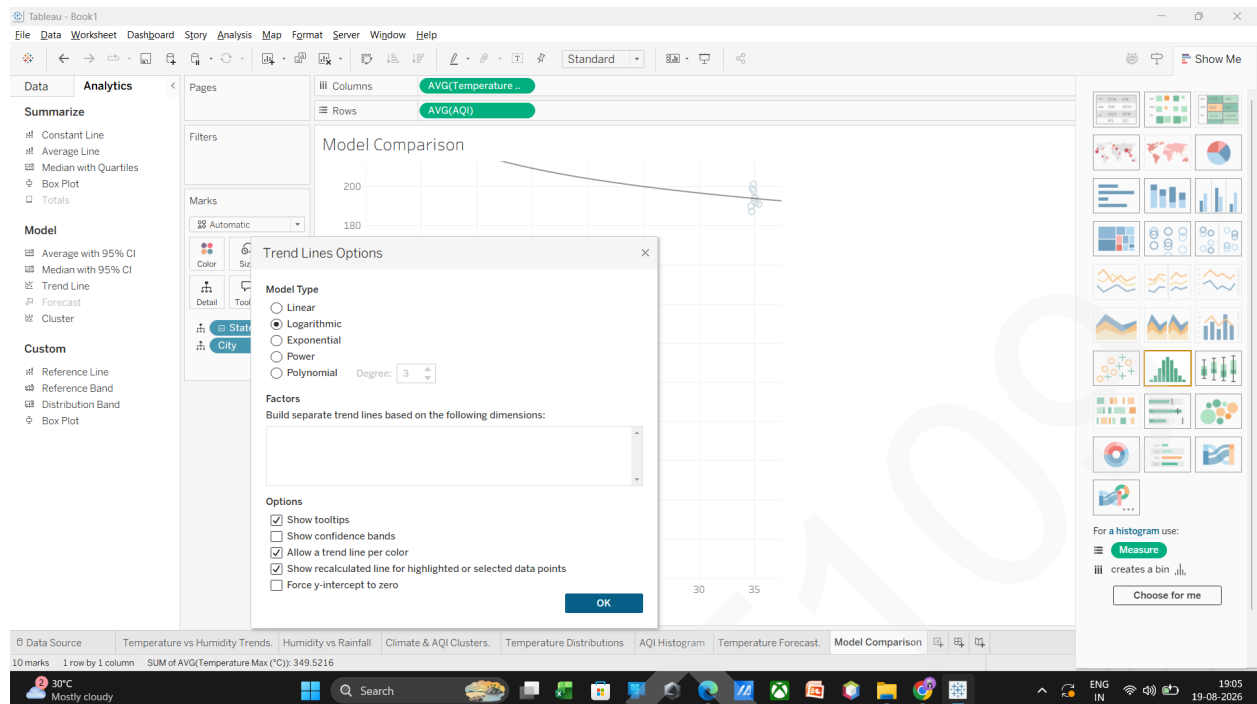


2. Add and Swap Statistical Models



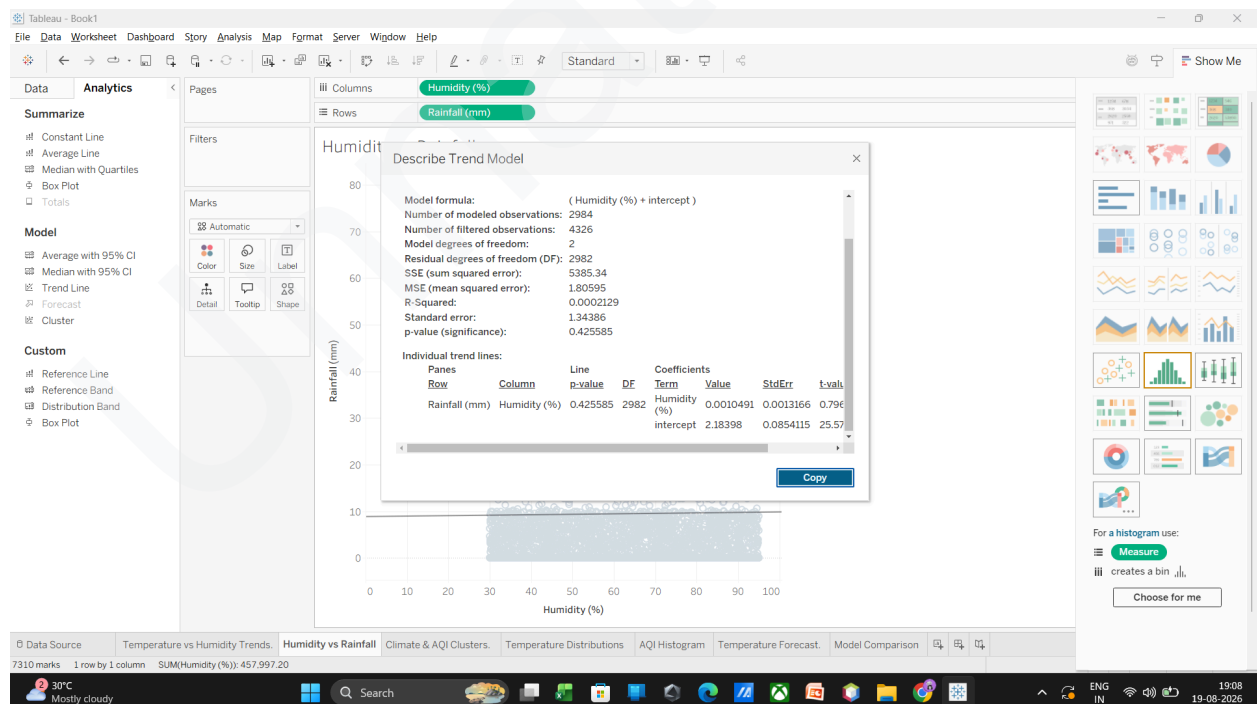
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G: Export and Interpret Statistical Results

1. Export the Statistical Model Description
2. Interpret the Key Output Metrics



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